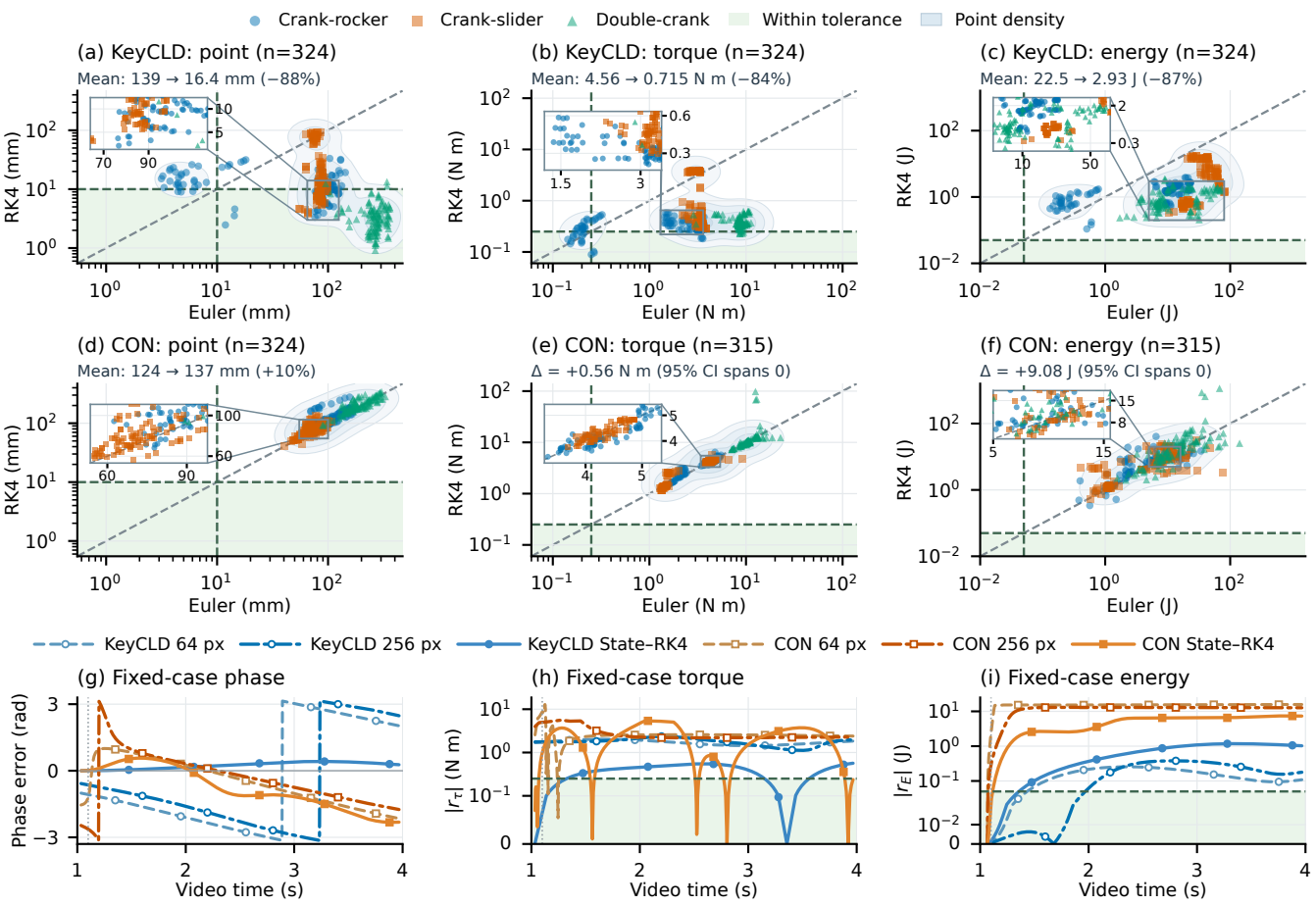

Crank-rocker
Crank-slider
Double-crank
Within tolerance
Point density
(a) KeyCLD: point (n=324)
Mean: 139 → 16.4 mm (−88%)
RK4 (mm)
Euler (mm)
(b) KeyCLD: torque (n=324)
Mean: 4.56 → 0.715 N m (−84%)
RK4 (N m)
Euler (N m)
(c) KeyCLD: energy (n=324)
Mean: 22.5 → 2.93 J (−87%)
RK4 (J)
Euler (J)
(d) CON: point (n=324)
Mean: 124 → 137 mm (+10%)
RK4 (mm)
Euler (mm)
(e) CON: torque (n=315)
Δ = +0.56 N m (95% CI spans 0)
RK4 (N m)
Euler (N m)
(f) CON: energy (n=315)
Δ = +9.08 J (95% CI spans 0)
RK4 (J)
Euler (J)
KeyCLD 64 px
KeyCLD 256 px
KeyCLD State–RK4
CON 64 px
CON 256 px
CON State–RK4
(g) Fixed-case phase
Phase error (rad)
Video time (s)
(h) Fixed-case torque
$|r_\tau|$ (N m)
Video time (s)
(i) Fixed-case energy
$|r_E|$ (J)
Video time (s)